

A Memo About Memos

Alejandro Revilla (@apr)

2024-07-24

Contents

A Memo About Memos	2
Required tools	3
TexLive	3
Pandoc	3
Mermaid-filter	3
M4	3
Pandoc template	4
Table-width filter	4
Pandoc LaTeX Environment	4
The memo script	6
Frontmatter	6
Bibliography	7
Revision History	8
References	10

A Memo About Memos

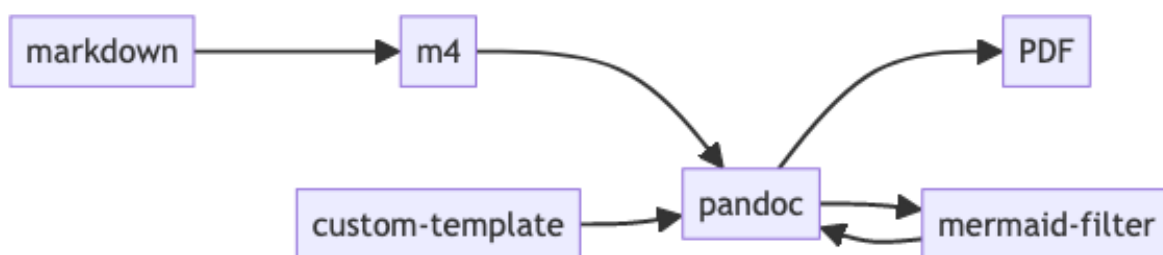
I am such a weak writer that I can't afford to add bad typography on top of my already flawed writing. I'm so challenged when it comes to aesthetics that, whenever I use a WYSIWYG editor—and I've used many since the early PageMaker days—I spend more time figuring out fonts and paragraph alignment than actually writing. Because I occasionally get to read beautifully typeset documents—and terrible ones most of the time—I can tell the difference and appreciate subtle details such as appropriate hyphenation, justification, kerning, leading, and the use of ligatures.

I write a lot of memos for different audiences, including technical, executive, internal, customer proposals, design analysis, and more. Over the years, I have used various tools to write these memos, such as YODL [1], $\text{T}_{\text{E}}\text{X}$ [2], $\text{\LaTeX}$ [3], DocBook [4], DITA [5], AsciiDoc [6], AsciiDoctor [7], and of course, Markdown¹. After decades of peregrination through different toolchains and editors, including high-end XML editors, nothing beats `vi` (or `emacs` if that's your religion) for fast editing, `Markdown` for ease of use, and $\text{\LaTeX}$ for perfect typesetting.

While the awesome `pandoc` is the easiest way to convert `Markdown` to a properly typeset `PDF`, the default output, though better than that of many other Markdown editors, is great for occasional memos but doesn't look professional. I've been trying to write a nice `pandoc` template myself without much success, but I recently came across a beautiful one: Eisvogel.²

This document could end right here: write Markdown and use 'pandoc' with the 'Eisvogel' template to produce beautiful documents. Period.

However, I believe I can contribute a small addition to the toolchain that will enhance your writing experience and content reuse.



¹ Disregard the bibliographic references and bookmarks like this; I'm adding them just to show how they look.

² <https://github.com/Wandmalfarbe/pandoc-latex-template>

Required tools

TexLive

T_EXLive is a comprehensive distribution of the T_EX typesetting system, offering a wide range of T_EX-related software, packages, fonts, and utilities. Maintained by the T_EX Users Group, it supports multiple operating systems, including Linux, macOS, and Windows. The live distribution can be found at: <https://www.tug.org/texlive/>

Pandoc

Pandoc is a powerful document converter that supports over a dozen input formats and more than thirty output formats, including Markdown, HTML, L^AT_EX, PDF, and Word, making it a perfect fit for our memo toolchain and can be found at <https://pandoc.org/>

Mermaid-filter

Mermaid is a versatile tool for creating diagrams and charts from simple text descriptions. It generates flowcharts, sequence diagrams, Gantt charts, and more using an easy-to-learn syntax. Popular in technical documentation and project management, Mermaid integrates seamlessly with various platforms, enhancing documents with clear and visually appealing diagrams. We need the `mermaid-filter` package that gets called from `pandoc`, and can be installed with the command:

```
npm i -g mermaid-filter
```

M4

The `m4` macro processor, designed by Brian Kernighan and Dennis Ritchie and originally developed as part of the Unix operating system in the 1970s, is not technically required in this toolchain but is a perfect fit for splitting large documents into smaller pieces, reusing content snippets, and providing standard constants such as the Git revision of the document, date, time, filename, and more. Key features of `m4` include its ability to define macros, perform arithmetic operations, manipulate text with ease, and call the operating system³. It also supports file inclusion, conditional statements, and looping constructs.

As an example, to show the revision of the document, you can insert `__REVISION__`, which will be expanded in this case to `2b4859c*`. The same goes for other variables such as `__DATE__`, `__TIME__`, `__FILENAME__`, `__BRANCH__` and others that you may want to add to the `memo`

³ This poses a security risk when using third-party Markdown sources.

script. If you're on Unix, `m4` is probably installed, verify that with the command `type m4`. On Windows, you can install it using Cygwin, or the Windows Subsystem for Linux (WSL).

Pandoc template

The Eisvogel template, located at <https://github.com/Wandmalfarbe/pandoc-latex-template>, has a fantastic set of examples and provides detailed instructions on how to install it. On macOS, with `pandoc` installed via `brew`, the location is `$HOME/.local/share/pandoc/templates` under the name `custom_eisvogel.latex`. It can, of course, be the pristine `eisvogel.latex` script. I suggest calling it `custom_eisvogel.latex` in case you want to add some additional $\text{\LaTeX}$ commands or configuration.

In that `pandoc/templates` directory, you may also want to add `ieee-with-url.csl` from <https://github.com/citation-style-language/styles/blob/master/ieee-with-url.csl> or another citation style of your choice.

Table-width filter

Pipe-table layout in `pandoc` is famously indirect: a table renders at natural width until any of its source lines exceeds `--columns` (72 by default), at which point it switches to full text width with relative column widths taken from the dash counts in the separator row. In other words, trailing spaces and dash runs are load-bearing. The `filters/table-widths.lua` filter in this repository removes that dependency by assigning every table explicit, content-proportional column widths; a table wrapped in a fenced div `::: { .natural-width }` keeps pandoc's default rendering. `memo` applies every `*.lua` filter found in `$HOME/.local/share/pandoc/filters/`, in name order (prefix filters with a number to control sequencing), and behaves exactly as before when the directory is absent. To enable this one:

```
1 mkdir -p $HOME/.local/share/pandoc/filters
2 cp filters/table-widths.lua $HOME/.local/share/pandoc/filters/
```

Pandoc LaTeX Environment



This is optional, but can be used to print tips like this.

If Pandoc LaTeX environment is available, and you include `awesomebox` in your `frontmatter` like we do in the sample one provided here, you can use `::: tip` blocks as well as `::: note`, warning caution and important admonitions.

Installation is easy, just:

```
1 pipx install pandoc-latex-environment
```

The memo script

The memo script can be located at <https://github.com/ar/memo/blob/main/bin/memo> and has the following options:

```
Usage: memo [options] file.md [single|double]

Options:
  --info           Verbose logging to <file>_processing.log instead of /dev/null.
  --tex           Force TeX intermediate even if no glossary/nomenclature.

  --glossary       Build LaTeX glossaries (makeglossaries + two extra XeLaTeX runs).
  --nomenclature   Build LaTeX nomenclature (makeindex + two extra XeLaTeX runs).

  --open           Automatically open the generated PDF (macOS, or Linux with xdg-
open/open).
  --yoink          Send the generated PDF to Yoink (macOS only).

  --keep           Keep all temporary / intermediate files (no cleanup at exit).
  --clean          Delete temporary / intermediate files and exit without building.
  --template PATH  Path to the root pandoc template name (it defaults to 'custom_eisvogel' from the default pa

  -DNAME=VALUE     m4 define passed through to preprocessing (repeatable).
```

If called without options, it converts the Markdown document directly to PDF. For documents that include glossary or nomenclature entries—both of which require multiple LaTeX passes—the `--tex` switch can be used to generate an intermediate `.tex` file. This can then be processed manually, optionally using the `--glossary` and `--nomenclature` switches if those features are needed.

Frontmatter

This is a sample frontmatter for this document:

```
1 ---
2 title: A Memo About Memos
3 subject: Memo Typesetting
4 author: Alejandro Revilla (\@apr)
5 date: 2024-07-24
6 keywords: [Typography]
7 titlepage: true
8 # titlepage-logo: jpos.jpg
9 lang: "en"
10 toc: true
11 toc-own-page: true
12 footer-center: jPOS.org (2b4859c*)
13 titlepage-rule-color: "360049"
14 titlepage-text-color: "FFFFFF"
15 titlepage-rule-height: 0
16 titlepage-background: "backgrounds/background.pdf"
```

```
17 footnotes-pretty: true
18 header-includes:
19   - \usepackage{multicol}
20   - \usepackage{mdframed}
21   - \usepackage{graphicx}
22   - \usepackage{xcolor}
23   - \usepackage{pgfplots}
24   - \usepackage{amsmath}
25   - \pgfplotsset{compat=1.17}
26   - \usepackage{longtable}
27   - \usepackage{fontawesome5}
28   - \newcommand{\user}{\faIcon{user}}
29   - \usepackage{awesomebox}
30
31 pandoc-latex-environment:
32   noteblock: [note]
33   tipblock: [tip]
34   warningblock: [warning]
35   cautionblock: [caution]
36   importantblock: [important]
37
38
39
40 ---
```

Please note that the front matter has been included in the document. Refer to memomemo.md for details.

Note that some of the added packages are used in my custom template, just as a PoC of different features. You can safely remove that `header-includes` section.

Bibliography

You can add `.bib` file, with the same name as your main `.md` file with the following format:

```
1 @article{kubat1997yodl,
2   title = {YODL or Yet Oneother Document Language},
3   author = {Kubat, Karel},
4   journal = {Linux Journal},
5   year = {1997},
6   month = {November},
7   number = {2089},
8   url = {https://www.linuxjournal.com/article/2089}
9 }
10
11 @manual{docbook,
12   title = {DocBook 5.0: The Transition Guide},
13   author = {{OASIS DocBook Technical Committee}},
14   year = {2010},
```



```

15   note = {Available online at \url{https://www.docbook.org/tdg5/en/html
16         /docbook.html}},
17   url = {https://www.docbook.org/tdg5/en/html/docbook.html}
18 }
19 @manual{dita,
20   title = {DITA 1.3 Specification},
21   author = {{OASIS DITA Technical Committee}},
22   year = {2015},
23   note = {Available online at \url{https://docs.oasis-open.org/dita/
24         dita/v1.3/os/}},
25   url = {https://docs.oasis-open.org/dita/dita/v1.3/os/}

```

Revision History

Date	Message	Author	Hash
2026-08-08	Merge pull request #7 from alcarraz/open-support-linux	A.Revilla	2b4859c
2026-08-08	Merge pull request #8 from alcarraz/fix/mv-same-on-linux	A.Revilla	de4cd0a
2026-08-08	Merge pull request #10 from ar/fix-early-errors	A.Revilla	8a77516
2026-08-08	Fail cleanly on a missing input file; run on older pandoc	A.Revilla	9dfaee6
2026-08-08	Merge pull request #9 from ar/table-widths-filter	A.Revilla	0cab463
2026-08-08	Deterministic table column widths via a pandoc Lua filter	A.Revilla	3ecf925
2026-07-13	Fix mv self-move failure on Linux in TeX build path	A.Alcarraz	88582f8
2026-07-10	Support -open on Linux via xdg-open/open	A.Alcarraz	63104db
2026-06-17	Merge pull request #6 from barspi/enhancements-memo-script	A.Revilla	b6c633f
2026-06-17	memo script: delaying pandoc command dependence until after -clean option is processed	B.Spinak	1562bad
2026-06-16	memo script enhancements	B.Spinak	aad7a18
2026-06-16	Regenerate PDF	A.Revilla	aa72ae3
2026-05-04	use \onecolumn even if we are using 'double' class	A.Revilla	50cc2a4
2026-04-10	remove duplicate titlepage-rule-color	A.Revilla	dcabfff
2026-04-01	Merge pull request #5 from alcarraz/main	A.Revilla	170e817
2026-04-01	fix for MacOS	A.Alcarraz	eb4ab3a
2026-03-31	Add repo root to include paths	A.Alcarraz	bd83676
2026-02-02	Merge pull request #4 from gregorioosorio/patch-1	A.Revilla	c4c5d04
2026-01-30	Change installation command to use pipx	G.Osorio	f633224
2025-10-05	chore: rebuild pdf	A.Revilla	4889288
2025-10-05	feat: add support for single file log	A.Revilla	339e0d9
2025-10-05	feat: add awesomebox config	A.Revilla	4ab727d
2025-10-03	Add pandoc-latex-environment support - Kudos @mcampiglia !	A.Revilla	5da6319
2025-10-01	add some debug options	A.Revilla	37483c3
2025-04-26	Extract mermaid graphics with using intermediate TeX processing	A.Revilla	c099b78
2025-04-23	regenerated pdf based on 97e9b884	A.Revilla	c686ebd
2025-04-23	Add support for nomenclature and glossary processing	A.Revilla	97e9b88

Date	Message	Author	Hash
2024-11-01	Generate PDF	A.Revilla	ef22f21
2024-11-01	ignore temp *_preprocessed.md file	A.Revilla	ddaa199
2024-11-01	add revision history and better handling of dirty repos	A.Revilla	100cdf4
2024-08-29	Add link to source	A.Revilla	bcec67a
2024-08-29	Add PDF output	A.Revilla	7fb65ab
2024-08-29	A memo about memos	A.Revilla	748a83f
2024-08-29	Initial commit	A.Revilla	afed625

References

- [1] K. Kubat, “YODL or yet oneother document language,” *Linux Journal*, no. 2089, 1997, Available: <https://www.linuxjournal.com/article/2089>
- [2] D. E. Knuth, *The TeXbook*. in Computers & typesetting, volume a. Reading, Massachusetts: Addison-Wesley, 1984.
- [3] L. Lamport, *LaTeX: A document preparation system*, 2nd ed. Reading, Massachusetts: Addison-Wesley, 1994.
- [4] OASIS DocBook Technical Committee, *DocBook 5.0: The transition guide*. 2010. Available: <https://www.docbook.org/tdg5/en/html/docbook.html>
- [5] OASIS DITA Technical Committee, *DITA 1.3 specification*. 2015. Available: <https://docs.oasis-open.org/dita/dita/v1.3/os/>
- [6] E. Stuart, *AsciiDoc user guide*. 2002. Available: <http://www.methods.co.nz/asciidoc/>
- [7] D. Allen, *Asciidoctor documentation*. 2021. Available: <https://asciidoctor.org/docs/>